# Tau2-bench: preserving requested user actions

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## Recommendation and scope

Prioritize explicit target-state semantics for the simulator's Wi-Fi and roaming tools. The simulator-facing interface should support deliberate ON/OFF requests as well as inversion. Interpretation and persona remain the simulator's responsibility: it can still misunderstand, choose the wrong target, omit a step or use the original toggle. Reliable execution of a chosen action does not require perfect customer behavior. The second limitation below, autonomous repair, remains unresolved.

The simulator chooses speech and user-tool calls; the orchestrator routes them, the environment executes actions, and the evaluator checks outcomes. I focus on telecom because user-side tools expose the relationship between instructions, actions and outcomes; these cases establish failure mechanisms, not prevalence across domains. I inspected implementation, instructions, scoring and public conversations. Evidence comes from the February 26, 2026 gpt-5.2 run: 114 tasks, four trials each. A screen of 12 candidates led to two detailed cases. The recorded commit c604716 differs from this fork's implementation base cc1df47. Both cited tasks use environment assertions for final reward: successful resolution does not establish faithful customer behavior.

## Two limitations, with exact evidence

**A. Requested state becomes blind inversion.** Case A, simulation 17959bc5-43ff-4e42-8cd3-5899e244f3cf, concerns MMS failure while abroad, with missing SIM and broken messaging permissions (Hard persona). At turn 12 the agent requests roaming ON; turn 13 calls toggle_roaming({}); turn 14 reports OFF. At turn 42 the agent says to keep Wi-Fi OFF; turns 43-44 instead enable it, and turns 45-46 undo that change. Expected: preserve an already-correct state or clarify confusion. Observed: two opposite-state mutations. Likely mechanism: mapping an on/off instruction to a no-argument inversion tool without checking state. The trace establishes the mismatch, not the model's internal reasoning. Impact: avoidable disturbances and extra interactions can confound agent evaluation. Recorded reward is 0, but later agent troubleshooting also fails; these inversions are not established causes of final failure.

**B. Autonomous repair and unsupported action reporting.** Case B, simulation d49a1fb1-72a1-44d5-92b3-9f18524c7bb3, requires excellent cellular speed abroad without Wi-Fi or a plan change (Hard persona). Turn 10 requests roaming, an airplane-mode cycle and a conditional reboot, and asks for status. It does not request enabling mobile data. After reading data OFF, the simulator independently calls toggle_data at turn 17; turn 18 confirms ON. Turn 19 also reports an airplane-mode on/off cycle, although turns 13-14 show only one switch from ON to OFF. Expected: perform requested repairs, report actual actions, and ask about the remaining data setting. Likely cause: helpful goal completion and fluent narration overriding tool-use constraints already present in the user guidelines. Impact: the simulator may repair an agent omission or mislead subsequent diagnosis. Recorded reward is 1; a counterfactual is needed to attribute any score inflation.

## Compare two implementable approaches

**Prompt procedure:** add a short rule to the tool-enabled user guidelines: identify the requested change; inspect the relevant state; change only if needed; verify and report only executed actions. Permit necessary reads, but ask before unrequested repairs. This targets both findings without changing schemas. It is cheap to implement but remains probabilistic, adds reads/tokens, and may make low-skill personas unnaturally methodical or overcautious.

**Target-state tools (recommended):** add optional enabled=true/false to existing toggle_wifi and toggle_roaming. A matching state returns unchanged; otherwise existing mutation helpers execute the change. Omission/null retains legacy inversion. Two signatures, docstrings and guarded no-op branches are the minimal change. The internal state comparison implements the chosen target; it neither diagnoses the problem nor selects the correct target for the LLM. Read-only status tools remain available for explicit inspection. This directly supports deliberate actions while preserving mistakes and genuine toggles. Risks: omitted or wrong arguments, skipped reset sequences, and changed behavior from the expanded schema. Orchestrator and evaluator code remain unchanged, although altered actions can change later conversations and rewards. The patch does not enforce authorization or fix B. Evaluate prompt changes separately.

Prioritize targets because the interface mismatch admits a narrow, testable intervention. Preventing autonomous repair requires broader judgments about which actions an instruction authorizes. This is a priority based on evidence and tractability, not a claim that inversions are globally the most important limitation.

## Implementation and what has actually been verified

The fork implements target states in src/tau2/domains/telecom/user_tools.py. Eighteen new offline tests cover state combinations and retries, legacy calls, existing connection preservation, disconnect side effects, airplane-mode restrictions, provider permission separation, schema export and environment dispatch. All 70 telecom tests pass under Python 3.12 with UTF-8 mode; targeted lint and formatting checks pass. No model experiment has been run. These checks validate mechanics, not model adoption, human realism or reward improvement. Task data and evaluator are unchanged. Treat this as an experimental fork, not an official comparable benchmark result.

## Small baseline-versus-change experiment (proposed)

**Fixed conditions.** Run baseline cc1df47 and the same checkout plus this patch with identical tasks, initial states, persona, simulator model/version, agent model/version, prompts, decoding parameters, tool limits and evaluator. Only the two user-tool schemas and implementations differ. Archive exact configurations and outputs. Never compare the historical public score directly with a new patched run.

**Cases and budget.** Build 12 short fixtures: two tools x two initial states x target ON/OFF (eight), plus an explicit toggle cycle, an ambiguous request, an unrequested-repair temptation, and an airplane-mode restriction. Use the cited agent utterances where applicable, with labels fixed before generation. For historical-prefix fixtures, reconstruct the environment and replay preceding tool calls, checking state against recorded responses before resuming; discard mismatches. Run five matched seeds per fixture per arm (120 short continuations). Pair seeds for variance reduction, not as a promise of deterministic model outputs. Separately run six fixed telecom tasks, including the two cited task types and controls, for three matched trials per arm to inspect end-to-end consequences (36 episodes).

**Primary measurements.** Score the complete action sequence, not just its endpoint. For "keep OFF," OFF-to-ON-to-OFF is a deviation even though the final state is correct; for "turn OFF, then ON," both transitions must occur in order. Record final target attainment separately. Measure unrequested writes per opportunity, action-report contradictions per checked claim, extra calls and target-argument adoption. Controls check clarification on ambiguity and existing blocked-action behavior. Record task reward and task-specific environment assertions as secondary outcomes, not the simulator-quality objective.

**Variability and interpretation.** Report numerator/denominator and paired differences by case and seed, plus task-level results. Bootstrap over fixtures/tasks, keeping repetitions clustered; show raw counts because intervals will be wide at this size. A promising result reduces wrong-state mutations without increasing unauthorized writes or contradictions and without damaging controls. Predefine a zero-new-failure rule for deterministic safety restrictions. Any reward change must be traced to the changed interaction, with both improved and degraded episodes reviewed.

## Challenge the conclusion: fidelity before easier tasks

Give two blinded reviewers the intended persona, permissible knowledge, visible settings, agent request and action trace. Label instruction adherence separately from whether each deviation is plausible for that persona; a missed instruction is not automatically a simulator defect. Also assess scenario constraints and adequacy of agent guidance, then adjudicate disagreements. Zero simulator mistakes is not the objective. Our hypothesis is improved action precision without a more competent persona; error frequencies and intent remain unestablished. Case A's turns 50-53 already show correct check-before-action behavior, contradicting any universal failure claim.

I would revise the recommendation if reviewers find a material loss of persona fidelity, if the model rarely adopts the parameter, or if genuine procedures regress. In particular, "ensure ON" requires a final state, whereas "turn OFF, then ON" requires both transitions; score the sequence, not just its endpoint. In the unrequested-repair control, the missing instruction should prompt clarification rather than autonomous repair. Higher reward alone is insufficient: better evaluation means fewer interface-induced errors while preserving uncertainty and the agent's responsibility to diagnose and guide.

## Sources, fork and disclosure

Paper; public trajectories. Full task snapshots, exact original zero-based turns and provenance: Case A, Case B and review notes. Fork contains the implementation, tests and annotations. Codex assisted with analysis, extraction, code, tests and drafting; source turns and test claims were checked.